Weather, Interconnections, and Energy Prices

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Abstract

We build an auditable European electricity-price dataset and evaluate three XGBoost variants and real/shuffled graph attention models across 49 market identities. Real-graph GAT achieves 20.83 EUR/MWh test MAE, improving 3.32% over its shuffled control and 4.75% over recent-price XGBoost. Results are exploratory and conditional on unverified historical publication vintages.

1 Approach

At 12:00 Europe/Brussels on calendar day $D-1$, predict every hourly day-ahead price of delivery day D, including 23/25-hour daylight-saving days. We implement an independent dataset builder, per-zone XGBoost [1], and pooled hourly GATv2 models [2]. Directed graph edges are physical pairs observed during training; historical node lifetimes mask inactive identities. A shuffled control rewires non-self edges, preserving static in/out degrees without duplicates. Both GATs add self-loops, testing real topology under the same neural learner.

2 Experiments

Exact data and partitions. The master contains 5,887,954 zone-hours for 62 price-area identities, January 2016–October 2026. ENTSO-E prices, load forecasts and physical flows use releases v1.2. Additional sources are PECD/ERA5 temperature, anomaly and wind/solar capacity factors (2010–September 2026), and original CWE/CORE JAO exports (May 2015–October 2026). Weather is reanalysis, not archived forecasts; its requested v1.1 release identity is unverified. Missing values and negative prices are retained, and DE_AT_LU remains separate from DE_LU. Forty-nine identities have valid targets in fit, validation and test; the other 13 remain explicitly unscored, while all 62 can provide dated graph context.

Fit on January 1, 2016–March 31, 2024 (pretrain + train); validate on April 1, 2024–March 31, 2025; test on April 1, 2025–September 30, 2026, using Brussels delivery dates. October stays unassigned. MAE and RMSE are measured after inverse transformation, in EUR/MWh. Weekly persistence copies the same identity's price 168 elapsed hours earlier. Targets are never filled or filtered for feature completeness; neural preprocessing is fitted on training-period features only.

Five models. Recent-price XGBoost uses local hour, weekday, month and guarded 24/168-hour price lags. Richer XGBoost adds load forecast, four realised weather features at lag 48, and three lag-48 JAO summaries: presolved constraint count, minimum RAM ratio and domain presence. Neighbour XGBoost adds incoming neighbours' recent-price means and finite-value counts. Both GATs use the richer node inputs; they share parameters across zones, whereas XGBoost fits per zone. Missing inputs remain NaN for trees; neural models use training medians/scaling and missingness indicators. No same-day realised flow or target price is supplied as a predictor.

All models fit $z = \operatorname{asinh}(p/50)$ with L1 loss and invert predictions as $50\sinh(\hat{z})$. XGBoost uses depth 6, at most 600 trees, learning rate 0.05, row/column sampling 0.8 and 50-round validation patience. GAT uses two layers, 64 hidden channels, four heads, dropout 0.1, Adam at 0.001, batch size 64, at most 30 epochs and patience five. Seed is 224; stopping uses transformed validation MAE. CPU XGBoost and ROCm GAT runs completed in about six minutes total.

Full evaluation. Table 1 uses identical 428,213 validation and 633,885 test zone-hours with valid targets and finite weekly history. Country summaries pool bidding-zone errors, rather than creating national average prices. Combined markets and unresolved ENTSO-E identifiers remain distinct.

Method (49 market identities)	Validation MAE	Test MAE
Weekly persistence	30.18	29.13
Recent-price XGBoost	21.18	21.87
Richer XGBoost	21.28	22.13
Neighbour XGBoost	20.83	21.87
Real-graph GAT	20.81	20.83
Shuffled-graph GAT	21.23	21.54

Table 1: Matched MAE (EUR/MWh), lower is better. One seed, 49 market identities.

Real-graph GAT improves test MAE by 3.32% over shuffled GAT, 4.75% over recent-price XGBoost, and 28.51% over persistence. The richer feature bundle does not improve aggregate XGBoost test MAE; neighbour inputs improve validation but nearly match the recent-price test baseline. These group additions do not isolate individual weather/load/JAO effects. The full target population, including hours without persistence, has 428,427 validation and 634,578 test hours. On all test targets, real GAT RMSE is 34.40 versus 39.95 for recent-price XGBoost. Real GAT selects epoch 28 of 30; shuffled GAT selects epoch 21 and stops at 26. Both real and shuffled runs use about 0.185 GiB peak tensor allocation, excluding ROCm context. All saved predictions have unique keys and matching target masks; independent recomputation agrees with reported MAE.

Earlier pipeline diagnostics. We reproduced a teammate's richer XGBoost pipeline at commit `757da793`: 43-zone test MAE 21.25 versus published 21.24. Its legacy table, geography and inputs differ, so this is a separate cohort from Table 1. In a fixed France learner, adding prior-day auction prices reduces test MAE from 32.09 to 22.20 (30.8%). Validation improvement is 11.16 EUR/MWh (95% paired calendar-week bootstrap interval [8.55, 14.13], 2,000 resamples conditional on the fitted models). Masking post-origin realised weather gives 22.06 test MAE versus 22.04 when supplied; such weather does not explain most of that France gain. These diagnostics motivated the stronger recent-price baseline.

Timing and limitations. Previous-day auction prices are normally published on $D-2$ [3]; delivery time is not publication time. We guard elapsed lag 24 against reaching the target delivery date on the autumn transition. Weather, load and JAO historical revision/publication vintages remain unverified: lag 48 alone cannot establish availability, especially for retrospective ERA5 [4]. Scores are conditional benchmarks, not certified operational forecasts. Training-period topology is static; dated lifetime masks do not reconstruct every commissioning date. One seed and previous exploratory test inspection prevent a confirmatory significance claim. Real/shuffled GAT is the matched topology control; GAT versus per-zone XGBoost also changes parameter sharing. We have not evaluated detailed constraint nodes.

3 Future work

Repeat across seeds and assess paired loss uncertainty; compare pooled tabular models to separate topology gains from parameter sharing. Obtain archived weather forecasts and publication-aware load/JAO inputs. Then evaluate temporal encoders and detailed constraint nodes against the established recent-price and graph baselines, keeping physical flows, scheduled exchanges and MaxBex distinct.

References

[1] T. Chen and C. Guestrin. XGBoost: A scalable tree boosting system. *KDD*, 2016. arXiv:1603.02754.

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[3] Nord Pool. Day-ahead auction timeline. Accessed November 2026.

[4] ECMWF. ERA5 global atmospheric reanalysis. Accessed November 2026.